

Data Analytics Capstone

**Machine Learning-Based One-Hour-Ahead PM2.5 Forecasting Using Environmental
and Meteorological Data**

Final Report

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Abstract

Urban air pollution poses significant environmental and public health challenges, particularly in densely populated cities where elevated concentrations of fine particulate matter (PM_{2.5}) can adversely affect human health. Accurate short-term forecasting of PM_{2.5} can support air-quality monitoring, early-warning systems, and environmental decision-making. This study developed and compared eight supervised machine learning regression models for one-hour-ahead PM_{2.5} forecasting using the Beijing Multi-Site Air Quality Dataset from the UCI Machine Learning Repository. The dataset contains 420,768 hourly observations from 12 monitoring stations in Beijing, China, covering March 2013 to February 2017. The unit of analysis was an hourly station-level observation. After preprocessing, missing-value treatment, and construction of the one-hour-ahead target, 409,158 observations were retained.

The models were Linear Regression, Decision Tree, Random Forest, XGBoost, Gradient Boosting, AdaBoost, LightGBM, and CatBoost. The workflow was implemented in Python using Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, CatBoost, and SHAP. CPU parallel processing was used for Random Forest, while CUDA-based GPU acceleration was used for XGBoost. Performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 with a chronological 80:20 train-test split. LightGBM emerged as the leading model based on testing R^2 and achieved the lowest testing MAE and RMSE, with MAE of 9.67 $\mu\text{g}/\text{m}^3$, RMSE of 18.26 $\mu\text{g}/\text{m}^3$, and R^2 of 0.9522. Compared with the Linear Regression baseline, LightGBM reduced testing MAE by 4.9% and RMSE by 4.3%. Five-fold expanding-window cross-validation produced mean MAE of 11.98 $\mu\text{g}/\text{m}^3$, RMSE of 22.43 $\mu\text{g}/\text{m}^3$, and R^2 of 0.9166.

Model interpretability was assessed using gain-based feature importance and SHAP (SHapley Additive exPlanations). Current-hour PM_{2.5} was the dominant predictor,

accounting for approximately 98.17% of gain-based feature importance, while PM10, wind speed, dew point temperature, temperature, and hour provided additional predictive information. The framework could be integrated into air-quality monitoring systems to provide environmental agencies with one-hour-ahead PM2.5 forecasts and support timely pollution advisories and early-warning decisions.

Keywords: PM2.5 forecasting; Air quality forecasting; Machine learning; LightGBM; Time-series forecasting; SHAP; Environmental analytics; Beijing Multi-Site Air Quality Dataset.

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1. Introduction

1.1 Background and Context

Air pollution has become one of the most significant environmental and public health challenges worldwide, particularly in rapidly growing and densely populated urban areas. Exposure to air pollution has been linked to a range of health consequences, including respiratory and cardiovascular conditions, premature mortality, and reduced life expectancy (World Health Organization [WHO], 2021; Pope & Dockery, 2006). Consequently, improving air-quality monitoring and forecasting has become an important priority for governments, environmental agencies, and public health organizations (WHO, 2021).

Among the various atmospheric pollutants, fine particulate matter (PM_{2.5}) is of particular concern because its small particle size allows it to penetrate deeply into the respiratory system. Research has established important associations between long-term exposure to fine particulate matter and adverse health outcomes, highlighting the importance of monitoring and managing PM_{2.5} concentrations (Pope & Dockery, 2006). Accurate short-term forecasting of PM_{2.5} can complement existing monitoring systems by providing information about expected near-future pollution levels and supporting timely air-quality assessment and decision-making.

PM_{2.5} concentrations are influenced by complex interactions among multiple environmental factors. The predictor set includes concentrations of other pollutants, such as PM₁₀, sulfur dioxide (SO₂), nitrogen dioxide (NO₂), carbon monoxide (CO), and ozone (O₃), together with meteorological measurements covering temperature, atmospheric pressure, dew point temperature, rainfall, wind speed, and wind direction. Temporal patterns and differences among monitoring locations can also contribute to variations in observed concentrations. The combination of these factors creates a forecasting problem involving potentially nonlinear relationships and interactions among multiple predictors.

Machine learning provides a suitable framework for modelling such complex relationships because supervised learning algorithms can learn predictive patterns from historical environmental observations (Hastie et al., 2021). Ensemble learning approaches, including Random Forest, Gradient Boosting, XGBoost, LightGBM, and CatBoost, have been developed to improve predictive performance by capturing nonlinear relationships and interactions among predictor variables (Breiman, 2001; Friedman, 2001; Chen & Guestrin, 2016; Ke et al., 2017; Prokhorenkova et al., 2018).

The present study focuses on one-hour-ahead PM_{2.5} forecasting using the Beijing Multi-Site Air Quality Dataset obtained from the UCI Machine Learning Repository (Chen, 2017). The dataset contains 420,768 hourly observations from 12 monitoring stations in Beijing, China, covering the period from 1 March 2013 to 28 February 2017 (Chen, 2017). It includes measurements of major air pollutants, meteorological variables, temporal information, wind direction, and monitoring-station identifiers, providing a comprehensive basis for developing and evaluating short-term PM_{2.5} forecasting models.

The study develops and compares eight supervised machine learning regression models -Linear Regression, Decision Tree Regression, Random Forest Regression, Gradient Boosting Regression, AdaBoost Regression, XGBoost Regression, LightGBM Regression, and CatBoost Regression. Model performance is evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2), while chronological evaluation and expanding-window time-series cross-validation are used to assess predictive performance and temporal robustness. Model interpretability is further examined using feature-importance analysis and SHAP (SHapley Additive exPlanations), providing insight into the predictors contributing to the selected model's one-hour-ahead forecasts (Lundberg & Lee, 2017).

Overall, the study integrates machine learning forecasting, chronological evaluation, and model explainability for one-hour-ahead PM_{2.5} prediction.

1.2 Problem Statement

Air pollution continues to present substantial environmental and public health challenges, particularly in densely populated urban areas where elevated PM_{2.5} concentrations can adversely affect human health. Although air-quality monitoring networks provide valuable measurements of current pollutant concentrations, they do not indicate how pollution levels are likely to change in the immediate future. Accurate short-term forecasting can therefore complement conventional monitoring by providing advance information about potential changes in air quality and supporting timely environmental and public health decision-making (WHO, 2021).

Accordingly, this study addresses the problem of accurately forecasting hourly PM_{2.5} concentration one hour ahead using current-hour environmental information. The dependent variable is the next-hour PM_{2.5} concentration, while the independent variables include current-hour pollutant measurements (PM_{2.5}, PM₁₀, SO₂, NO₂, CO, and O₃), meteorological variables (temperature, atmospheric pressure, dew point temperature, rainfall, wind speed, and wind direction), temporal variables, and monitoring-station information obtained from the Beijing Multi-Site Air Quality Dataset (Chen, 2017). The dataset comprises 420,768 hourly observations from 12 monitoring stations in Beijing, China, covering the period from 1 March 2013 to 28 February 2017 (Chen, 2017).

The analytical challenge is to identify the most accurate and temporally robust regression approach for one-hour-ahead PM_{2.5} forecasting while preserving chronological structure. Model performance is compared using MAE, RMSE, and R², followed by expanding-window validation of the leading model.

1.3 Purpose of the Study

The purpose of this study was to develop, evaluate, and compare supervised machine learning regression models for forecasting PM_{2.5} concentration one hour ahead using pollutant, meteorological, temporal, wind-direction, and monitoring-station variables obtained from the Beijing Multi-Site Air Quality Dataset (Chen, 2017). Specifically, the study evaluated Linear Regression, Decision Tree Regression, Random Forest Regression, Gradient Boosting Regression, XGBoost Regression, LightGBM Regression, CatBoost Regression, and AdaBoost Regression to identify the model that provided the strongest predictive performance. Model performance was assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2) within a chronological evaluation framework (Hastie et al., 2021; Pedregosa et al., 2011).

Temporal robustness was assessed using expanding-window cross-validation, while gain-based feature importance and SHAP were used to interpret the selected LightGBM model and identify influential predictors.

2. Scope and Objectives

2.1 Scope of the Study

This study focuses on the one-hour-ahead forecasting of PM_{2.5} concentration using the Beijing Multi-Site Air Quality Dataset. The analysis covers hourly observations collected from 12 monitoring stations in Beijing, China, between March 2013 and February 2017. The study considers air-pollution, meteorological, temporal, wind-direction, and monitoring-station variables available within the dataset as potential predictors of the subsequent hour's PM_{2.5} concentration.

The scope of the modelling component is limited to eight supervised machine learning regression algorithms: Linear Regression, Decision Tree Regression, Random Forest Regression, Gradient Boosting Regression, AdaBoost Regression, XGBoost Regression,

LightGBM Regression, and CatBoost Regression. Model performance is evaluated using MAE, RMSE, and R^2 through a chronological train-test framework, followed by expanding-window time-series cross-validation of the leading model.

The study also includes model interpretability analysis focused on the selected LightGBM model. Gain-based feature importance and SHAP analysis are used to examine the relative contribution of predictor variables to the model's one-hour-ahead PM_{2.5} forecasts.

The study is specifically concerned with short-term, one-hour-ahead prediction and does not evaluate longer forecasting horizons, real-time deployment, or forecasting performance in cities outside the geographical coverage of the dataset. The findings are therefore interpreted within the temporal, geographical, and variable scope of the Beijing Multi-Site Air Quality Dataset.

2.2 Research Objectives

The primary objective of this study was to develop and evaluate an accurate and reliable machine learning framework for forecasting PM_{2.5} concentration one hour ahead using pollutant, meteorological, temporal, wind-direction, and monitoring-station information.

The specific objectives were:

1. To examine the pollutant, meteorological, temporal, wind-direction, and monitoring-station variables associated with one-hour-ahead PM_{2.5} concentration.
2. To analyze the relationships between current-hour predictor variables and one-hour-ahead PM_{2.5} concentration using exploratory data analysis and statistical techniques.
3. To develop and compare eight supervised machine learning regression models for forecasting PM_{2.5} concentration one hour ahead.

4. To evaluate and compare the predictive performance of the developed models using MAE, RMSE, and R^2 within a chronological train-test evaluation framework.
5. To assess the temporal robustness and generalization of the selected model using expanding-window time-series cross-validation.
6. To identify and interpret the variables contributing most strongly to the selected LightGBM model using gain-based feature importance and SHAP analysis.
7. To identify the most suitable machine learning model for one-hour-ahead PM_{2.5} forecasting based on predictive performance and discuss its potential application to short-term air-quality monitoring and environmental decision-making.

2.3 Research Questions

The study addressed the following research questions:

- **RQ1.** Which current-hour pollutant and meteorological variables are most strongly associated with PM_{2.5} concentration one hour ahead?
- **RQ2.** How accurately can machine learning regression models forecast PM_{2.5} concentration one hour ahead using current-hour pollutant, meteorological, temporal, wind-direction, and monitoring-station information?
- **RQ3.** Which machine learning regression model provides the best predictive performance for one-hour-ahead PM_{2.5} forecasting?
- **RQ4.** Which predictor variables contribute most strongly to the forecasts of the selected machine learning model?

2.4 Null and Alternative Hypotheses

The hypotheses corresponding to each research question are presented below.

Statistical significance was evaluated at the **5% significance level ($\alpha = 0.05$)**.

Research Question 1 (RQ1): Which current-hour pollutant and meteorological variables are most strongly associated with PM_{2.5} concentration one hour ahead?

- **Null Hypothesis (H_{01}):** There is no statistically significant association between current-hour pollutant and meteorological variables and one-hour-ahead PM2.5 concentration.
- **Alternative Hypothesis (H_{11}):** At least one current-hour pollutant or meteorological variable has a statistically significant association with one-hour-ahead PM2.5 concentration.

Research Question 2 (RQ2): How accurately can machine learning regression models forecast PM2.5 concentration one hour ahead using current-hour pollutant, meteorological, temporal, wind-direction, and monitoring-station information?

- **Null Hypothesis (H_{02}):** Machine learning regression models do not provide accurate predictive performance for forecasting PM2.5 concentration one hour ahead using the selected predictor variables.
- **Alternative Hypothesis (H_{12}):** Machine learning regression models provide accurate predictive performance for forecasting PM2.5 concentration one hour ahead using the selected predictor variables.

Research Question 3 (RQ3): Which machine learning regression model provides the best predictive performance for one-hour-ahead PM2.5 forecasting?

- **Null Hypothesis (H_{03}):** There is no statistically significant difference in predictive performance among the evaluated machine learning regression models.
- **Alternative Hypothesis (H_{13}):** At least one evaluated machine learning regression model demonstrates significantly better predictive performance than the others.

Research Question 4 (RQ4): Which predictor variables contribute most strongly to the forecasts of the selected machine learning model?

- **Null Hypothesis (H_{04}):** None of the predictor variables makes a statistically significant contribution to the forecasts produced by the selected machine learning regression model.
- **Alternative Hypothesis (H_{14}):** At least one predictor variable makes a statistically significant contribution to the forecasts produced by the selected machine learning regression model.

Note. The hypotheses provide a statistical framework for addressing the research questions. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), coefficient of determination (R^2), and expanding-window time-series cross-validation, while predictor importance for the selected LightGBM model was assessed using gain-based feature importance and SHAP (SHapley Additive exPlanations).

2.5 Contributions and Expected Value

This study contributes by systematically comparing eight machine learning regression models for one-hour-ahead PM_{2.5} forecasting within a chronological framework, providing evidence on their relative predictive performance.

The study also incorporates expanding-window time-series cross-validation to assess the temporal robustness and generalization of the leading model beyond a single train-test split.

Gain-based feature importance and SHAP analysis add interpretability by identifying influential predictors. From a practical perspective, the framework may support short-term air-quality monitoring, early identification of changing pollution conditions, and data-driven decision-making.

2.6 Chapter Summary

This chapter defined the study scope, objectives, research questions, hypotheses, and expected contributions, establishing the chronological machine learning, temporal validation, and interpretability framework used in the analysis.

3. Literature Review

3.1 Introduction

This chapter reviews literature on PM_{2.5} forecasting, air-quality prediction, machine learning regression, ensemble learning, time-series evaluation, and interpretable machine learning, with particular attention to supervised methods, temporal validation, XAI, and SHAP (Lundberg & Lee, 2017). The review synthesizes methodological approaches, predictors, metrics, and findings to identify the research gap and establish the foundation for comparative modelling, temporal validation, and interpretability.

3.2 Literature Review Matrix

Table 3.2

Summary of Key Literature Relevant to PM_{2.5} Forecasting

#	Author (Year)	Domain / Context	Dataset / Setting	Method(s)	Key Findings / Contribution	How it Supports RQ(s) / Project Decisions (RQ Linkage)
1	Chen (2017)	Beijing air-quality monitoring	Beijing Multi-Site Air Quality Dataset; 12 monitoring stations, hourly observations	Air-quality data collection and monitoring	Provides the air-quality dataset containing pollutant and meteorological variables used for empirical analysis.	Supports RQ1–RQ4 and the overall forecasting study.
2	Zhang et al. (2017)	Beijing PM _{2.5} analysis	Beijing air-quality and meteorological data	PM _{2.5} and weather analysis	Examined PM _{2.5} severity and relationships with meteorological conditions and seasonal effects.	Supports RQ1 and the selection of meteorological predictors.
3	Duan et al. (2023)	PM _{2.5} forecasting	Six Chinese urban agglomerations	Comparative machine learning using meteorological data	Demonstrated the applicability of machine learning methods for PM _{2.5} prediction and highlighted the relevance of meteorological variables.	Supports RQ1 and RQ2, particularly predictor selection and comparative model evaluation.
4	Haseeb et al. (2024)	PM _{2.5} forecasting	Air-quality and meteorological observations	Comparative machine learning	Showed the value of combining chemical and meteorological indicators for PM _{2.5} prediction.	Supports RQ1 and RQ2, particularly predictor selection and model comparison.
5	Breiman (2001)	Machine learning / ensemble prediction	General machine-learning applications	Random Forest	Established ensemble decision-tree modelling for improved predictive performance.	Supports the inclusion of Random Forest for RQ2 and RQ3.
6	Chen & Guestin (2016)	Machine learning / scalable prediction	General large-scale machine-learning applications	XGBoost	Developed an efficient and scalable gradient-boosting framework.	Supports the inclusion of XGBoost for RQ2 and RQ3.
7	Ke et al. (2017)	Machine learning / large-scale prediction	Large-scale structured datasets	LightGBM	Developed an efficient histogram-based gradient-boosting framework for large datasets.	Supports the inclusion of LightGBM for RQ2 and RQ3 and its evaluation as a candidate model.
8	Bergmeir et al. (2018)	Time-series forecasting	Autoregressive time-series data	Time-series cross-validation	Highlighted the importance of appropriate validation methods for dependent time-series data.	Supports chronological testing and expanding-window cross-validation for RQ2 and RQ3.
9	Lundberg & Lee (2017)	Explainable artificial intelligence	Machine-learning prediction models	SHAP	Established a unified framework for interpreting model predictions through feature contributions.	Supports SHAP-based interpretation for RQ4.
10	Yang et al. (2021)	PM _{2.5} forecasting / deep learning	Beijing air-quality data	LSTM, CNN, CNN-LSTM	Demonstrated the potential of deep learning for short-term PM _{2.5} forecasting.	Provides deep-learning context and future-extension evidence for short-term PM _{2.5} forecasting.

Table 3.2 (Continued)

#	Author (Year)	Domain / Context	Dataset / Setting	Method(s)	Key Findings / Contribution	How it Supports RQ(s) / Project Decisions (RQ Linkage)
11	Friedman (2001)	Machine learning / predictive modelling	General predictive modelling	Gradient Boosting	Established sequential gradient boosting as an approach for improving predictive accuracy through additive modelling.	Supports the inclusion of Gradient Boosting for RQ2 and RQ3.
12	Prokhorenkova et al. (2018)	Machine learning / categorical-data modelling	Structured datasets containing categorical variables	CatBoost	Developed a gradient-boosting method with effective handling of categorical features.	Supports the inclusion of CatBoost for RQ2 and RQ3, particularly given station and wind-direction information.
13	Hastie et al. (2021)	Statistical learning / predictive modelling	General machine-learning applications	Regression, tree-based methods, ensemble learning	Provides foundational principles for statistical learning, predictive modelling, model evaluation, and regularisation.	Provides methodological foundation for the modelling and evaluation approach in RQ2 and RQ3.
14	Pedregosa et al. (2011)	Machine learning / computational implementation	Python scientific-computing environment	Scikit-learn	Provided a widely used Python framework for machine-learning implementation, preprocessing, modelling, and evaluation.	Supports the computational implementation of ML models used for RQ2 and RQ3.
15	Pope & Dockery (2006)	Environmental health / air pollution	Epidemiological and air-pollution evidence	Review of health effects of fine particulate matter	Established evidence concerning adverse health effects associated with exposure to fine particulate matter.	Establishes the problem context and significance of PM2.5 forecasting.
16	World Health Organization (2021)	Air quality / public health	Global air-quality guidelines	Evidence-based guideline development	Established health-based air-quality guideline levels and the importance of controlling particulate and other pollutant exposure.	Establishes the public-health relevance and motivation for PM2.5 forecasting.
17	Vaswani et al. (2017)	Deep learning / sequence modelling	General sequence-modelling applications	Transformer architecture	Introduced the Transformer architecture based on self-attention mechanisms for modelling complex sequential relationships.	Provides theoretical context for potential future deep-learning extensions to temporal PM2.5 forecasting.
18	Yao et al. (2023)	Generative AI / agentic reasoning	Language-model reasoning and task environments	ReAct: reasoning and acting	Demonstrated an approach that combines language-model reasoning with actions and external information.	Provides context for potential future agentic decision-support and user-interaction extensions.
19	Molnar (2022)	Explainable machine learning	Machine-learning prediction models	Interpretable ML / feature-attribution methods	Provides practical foundations for interpreting machine-learning models and understanding feature contributions.	Supports the interpretability framework for RQ4.

3.3 Machine Learning for Air Quality Prediction

Air quality forecasting has evolved from traditional statistical modelling techniques to advanced machine learning approaches capable of capturing complex relationships among environmental variables. Conventional statistical models, such as linear regression and autoregressive methods, may be limited in their ability to represent nonlinear interactions between atmospheric pollutants and meteorological conditions. Machine learning algorithms provide an alternative approach by learning predictive patterns from historical environmental observations (Hastie et al., 2021).

The Beijing Multi-Site Air Quality Dataset introduced by Chen (2017) provides a comprehensive basis for air-quality forecasting research because it contains hourly observations of multiple air pollutants and meteorological variables across several monitoring stations. The availability of multiple environmental predictors and monitoring locations

enables researchers to develop and compare different machine learning algorithms for air-quality prediction under consistent observational conditions (Chen, 2017).

Among supervised learning techniques, ensemble-based algorithms provide effective approaches for modelling complex environmental relationships. Random Forest aggregates multiple decision trees to improve predictive stability and reduce variance (Breiman, 2001). Gradient Boosting builds models sequentially, with each learner attempting to improve upon the errors of previous learners (Friedman, 2001). More recent boosting algorithms, including XGBoost, LightGBM, and CatBoost, extend these approaches through improvements in computational efficiency, scalability, regularization, and the handling of structured and categorical data (Chen & Guestrin, 2016; Ke et al., 2017; Prokhorenkova et al., 2018).

These developments provide a strong methodological basis for applying machine learning to PM_{2.5} forecasting. Accordingly, the present study compares eight supervised machine learning regression models to evaluate their relative predictive performance for one-hour-ahead PM_{2.5} forecasting using pollutant, meteorological, temporal, wind-direction, and monitoring-station information.

3.4 Thematic Synthesis and Research Gap

The reviewed literature demonstrates that machine learning provides a valuable framework for short-term air-quality forecasting because of its ability to model complex relationships among pollutant, meteorological, and temporal variables. Ensemble-based methods, particularly tree-based and gradient-boosting algorithms, have become important approaches for environmental prediction because they can capture nonlinear relationships and interactions among predictors.

The literature also highlights the importance of selecting appropriate predictor variables for PM_{2.5} forecasting. Current pollutant concentrations, meteorological conditions, and temporal characteristics can provide complementary information for predicting short-

term changes in PM_{2.5} levels. However, predictive performance can vary across algorithms and datasets, making comparative evaluation important when selecting an appropriate forecasting model.

Another important theme is the need to account for the temporal nature of air-quality data. Conventional random train-test splitting may not adequately represent a forecasting situation because observations from different time periods can become mixed between training and testing datasets. Chronological evaluation and time-series cross-validation provide a more appropriate framework for assessing how well a model generalizes to future observations.

The literature further emphasizes the importance of model interpretability. While predictive accuracy is important, understanding which variables contribute to model predictions can provide additional insight into the forecasting process. SHAP provides a systematic approach for examining feature contributions and interpreting individual and global model behaviour (Lundberg & Lee, 2017).

Emerging deep learning and generative AI approaches also provide potential extensions to air-quality forecasting and decision support. Deep learning models such as LSTM and CNN-LSTM have demonstrated potential for short-term PM_{2.5} forecasting, including applications using Beijing air-quality data (Yang et al., 2021). Transformer architectures provide another approach for modelling complex sequential relationships (Vaswani et al., 2017). More recently, large language models (LLMs) and agentic approaches have created opportunities for natural-language interaction with analytical systems and coordination of tasks through external tools. ReAct, for example, combines language-model reasoning with actions (Yao et al., 2023). In the context of the present study, these approaches are considered potential future extensions for decision support and user interaction rather than components of the current forecasting model. The present study therefore focuses on

supervised machine learning and interpretable modelling while recognizing deep learning and emerging AI approaches as areas for future development.

Based on these themes, the present study addresses the need for an integrated short-term PM_{2.5} forecasting framework that combines comparative machine learning evaluation, chronological testing, temporal cross-validation, and model interpretability. The study therefore compares eight supervised machine learning regression models, evaluates them using a chronological train-test framework, assesses the selected LightGBM model using expanding-window time-series cross-validation, and examines its predictive drivers using gain-based feature importance and SHAP analysis. This integrated approach provides both predictive and interpretive evidence for one-hour-ahead PM_{2.5} forecasting.

4. Materials and Methods

4.1 Introduction

The methodology was designed to develop, evaluate, and interpret supervised machine learning regression models for one-hour-ahead PM_{2.5} forecasting using historical environmental and meteorological observations. The process included data acquisition, quality assessment, preprocessing, EDA, feature engineering, target construction, chronological splitting, model development, evaluation, temporal validation, and interpretation.

Eight regression models were compared using MAE, RMSE, and R^2 . The leading model was evaluated with expanding-window cross-validation, while gain-based feature importance and SHAP were used to interpret its predictions.

4.2 Research Design

This study employed a quantitative, predictive research design using supervised machine learning techniques. The primary objective was to forecast PM_{2.5} concentration one hour ahead by learning relationships between current-hour pollutant measurements,

meteorological variables, temporal information, wind-direction characteristics, and monitoring-station information.

A comparative modelling approach was adopted in which eight supervised machine learning regression algorithms were developed and evaluated using a common dataset and consistent performance measures. Model performance was assessed using a chronological train-test split to preserve the temporal ordering of observations. The leading model was subsequently evaluated using five-fold expanding-window time-series cross-validation to assess temporal robustness and generalization.

Following model selection, the LightGBM model was interpreted using gain-based feature importance and SHAP analysis. This enabled the study to examine both the relative importance of predictors and their contributions to the model's one-hour-ahead PM_{2.5} forecasts.

4.3 Data Source

The Beijing Multi-Site Air Quality Dataset was obtained from the UCI Machine Learning Repository (Chen, 2017). The dataset contains hourly air-quality and meteorological observations collected from 12 monitoring stations in Beijing, China, covering the period from 1 March 2013 to 28 February 2017. The dataset provides measurements of multiple air pollutants together with meteorological and temporal information, making it suitable for developing and evaluating short-term PM_{2.5} forecasting models.

In addition to pollutant concentrations, the dataset records several meteorological measurements, including temperature, atmospheric pressure, dew point temperature, rainfall, wind speed, and wind direction. It also contains temporal fields and station identifiers that support the analysis of changes in air quality across time and monitoring locations

For this study, the dataset was transformed to create the one-hour-ahead forecasting target (PM2.5_next_hour). The target represents the PM2.5 concentration at the subsequent hour within the same monitoring station. Consequently, the current-hour PM2.5 measurement was retained as a predictor because it represents information available at the time of forecasting.

The original dataset contained 420,768 hourly observations. Following preprocessing, target construction, and exclusion of observations without a valid one-hour-ahead target, 409,158 observations were retained for model development and evaluation.

4.4 Data Dictionary

Table 4.4

Data Dictionary for the Beijing Multi-Site Air Quality Dataset

Variable	Description	Data Type	Role in Study
No	Record identifier	Numeric	Identifier, excluded from modelling
year	Year of observation	Numeric	Temporal variable
month	Month of observation	Numeric	Temporal predictor
day	Day of month	Numeric	Temporal variable
hour	Hour of day	Numeric	Temporal predictor
PM2.5	Current-hour fine particulate matter concentration ($\mu\text{g}/\text{m}^3$)	Numeric	Pollutant predictor
PM10	Current-hour particulate matter ≤ 10 μm concentration ($\mu\text{g}/\text{m}^3$)	Numeric	Pollutant predictor
SO ₂	Current-hour sulfur dioxide concentration ($\mu\text{g}/\text{m}^3$)	Numeric	Pollutant predictor

Variable	Description	Data Type	Role in Study
NO ₂	Current-hour nitrogen dioxide concentration ($\mu\text{g}/\text{m}^3$)	Numeric	Pollutant predictor
CO	Current-hour carbon monoxide concentration ($\mu\text{g}/\text{m}^3$)	Numeric	Pollutant predictor
O ₃	Current-hour ozone concentration ($\mu\text{g}/\text{m}^3$)	Numeric	Pollutant predictor
TEMP	Temperature ($^{\circ}\text{C}$)	Numeric	Meteorological predictor
PRES	Atmospheric pressure (hPa)	Numeric	Meteorological predictor
DEWP	Dew point temperature ($^{\circ}\text{C}$)	Numeric	Meteorological predictor
RAIN	Precipitation (mm)	Numeric	Meteorological predictor
wd	Wind direction	Categorical	Meteorological predictor
WSPM	Wind speed (m/s)	Numeric	Meteorological predictor
station	Monitoring-station name	Categorical	Location-related predictor

Note. Variable names and descriptions are based on the Beijing Multi-Site Air Quality Dataset (Chen, 2017). For the forecasting analysis, the engineered target variable PM2.5_next_hour was created by shifting PM2.5 observations one hour forward within each monitoring station. Thus, the current-hour PM2.5 measurement was retained as a predictor, while PM2.5_next_hour served as the dependent variable. The record identifier (No) was excluded from modelling because it functions only as an identifier.

4.5 Inclusion and Exclusion Criteria

Inclusion Criteria

The study included hourly observations from all 12 monitoring stations in the Beijing Multi-Site Air Quality Dataset. Observations containing the pollutant, meteorological, temporal, wind-direction, and monitoring-station information required for model development were considered for analysis. Retaining observations across all monitoring stations preserved the spatial variability represented in the dataset.

Exclusion Criteria

Observations for which the one-hour-ahead target could not be generated were excluded from the forecasting dataset. Specifically, the final hourly observation within each monitoring station could not be assigned a subsequent-hour PM_{2.5} value and was therefore removed following target construction. Observations with missing values remaining after the preprocessing procedure were also excluded where required for model development.

No additional observations were removed solely on the basis of extreme values during preprocessing. The study retained the observed environmental variability represented in the dataset rather than applying an arbitrary outlier-removal threshold.

4.6 Data Preprocessing

The data preprocessing procedure was designed to prepare the observations for supervised machine learning while maintaining the chronological structure of the forecasting problem. The one-hour-ahead target variable, PM_{2.5}_next_hour, was constructed by shifting PM_{2.5} observations forward by one hour within each monitoring station. This ensured that the target represented the PM_{2.5} concentration at the subsequent hour while the predictor variables represented information available at the current hour.

Categorical variables, including wind direction (wd) and monitoring station (station), were converted into numerical representations through appropriate encoding. Observations for which the forecasting target could not be generated were removed. The resulting predictor

dataset was prepared for model development, with the record identifier (No) excluded because it served only as an identifier and did not provide meaningful predictive information.

The preprocessing procedure was applied before model development while maintaining the temporal ordering required for chronological evaluation. The resulting predictor matrix contained the variables used for the comparative modelling and subsequent interpretation of the selected LightGBM model.

Missing predictor values were forward-filled within monitoring station using previously available observations; the forecasting target was constructed before imputation so that target values were not generated from imputed PM2.5 observations.

4.7 Exploratory Data Analysis

Exploratory data analysis (EDA) was conducted to examine the characteristics of the dataset and identify relevant patterns before model development. Descriptive statistics, distributional analysis, correlation analysis, and temporal visualizations were used to examine pollutant concentrations, meteorological variables, and temporal patterns.

The EDA provided evidence of substantial variation in PM2.5 concentrations across observations and monitoring stations and revealed temporal patterns associated with different periods of the day and year. Current-hour PM2.5 demonstrated a strong relationship with the one-hour-ahead PM2.5 target, indicating the importance of recent pollution levels for short-term forecasting. Other pollutant and meteorological variables, including PM10, temperature, dew point temperature, and wind-related measures, also provided information relevant to the forecasting problem.

These exploratory findings supported the inclusion of pollutant, meteorological, temporal, wind-direction, and monitoring-station variables in the machine learning models. The EDA therefore provided an initial understanding of the data structure and informed the subsequent modelling and interpretation stages.

4.8 Experimental Workflow

The overall analytical workflow adopted in this study is presented in Figure 4.8. The workflow was designed to provide a systematic and reproducible framework for one-hour-ahead PM_{2.5} forecasting.

The process began with importing the required Python libraries and loading the Beijing Multi-Site Air Quality Dataset. Data quality was then assessed through missing-value analysis, duplicate verification, and examination of the observed data distributions. Exploratory data analysis was subsequently performed to investigate pollutant relationships, temporal patterns, and other characteristics of the dataset.

The PM_{2.5}_next_hour target was then constructed by shifting PM_{2.5} observations one hour forward within each monitoring station. Predictor variables were prepared through missing-value handling, categorical encoding, and feature preparation. The resulting dataset was divided chronologically into training and testing subsets to preserve the temporal structure of the forecasting problem.

Eight supervised machine learning regression models- Linear Regression, Decision Tree, Random Forest, XGBoost, Gradient Boosting, AdaBoost, LightGBM, and CatBoost - were developed and compared. Model performance was evaluated using MAE, RMSE, and R^2 on the chronological training and testing datasets.

The leading model, LightGBM, was subsequently evaluated using five-fold expanding-window time-series cross-validation to assess its temporal robustness. Finally, gain-based feature importance and SHAP analysis were applied to the selected LightGBM model to examine the relative importance and predictive contributions of the input variables.

Figure 4.8 illustrates the complete end-to-end workflow adopted for the one-hour-ahead PM2.5 forecasting analysis.

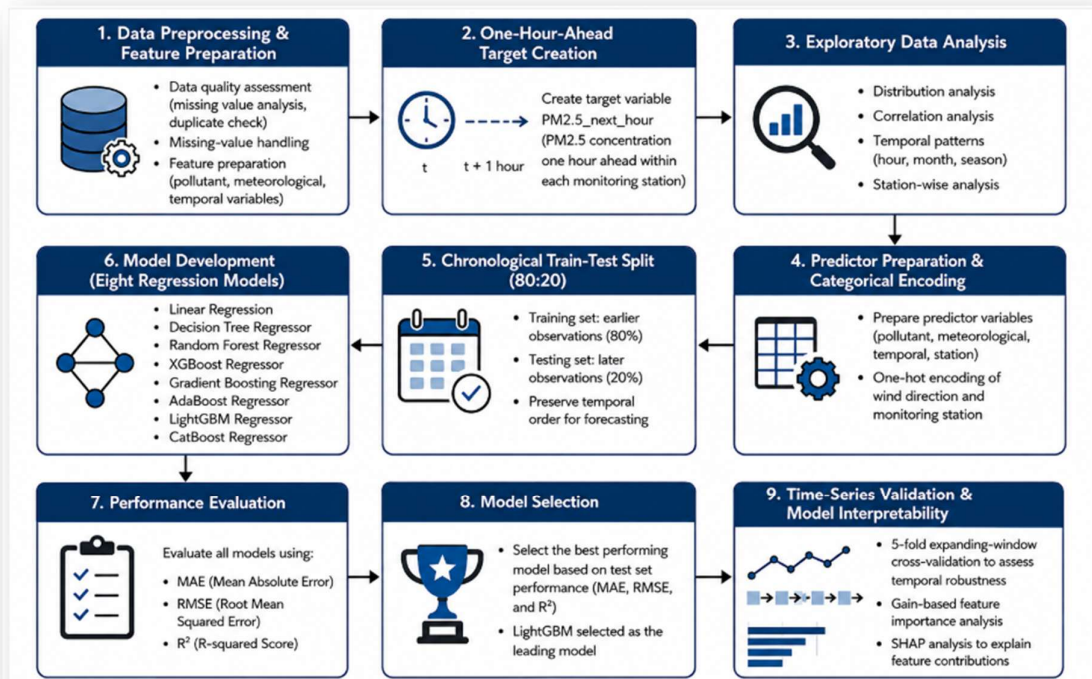


Figure 4.8

End-to-End Workflow for One-Hour-Ahead PM2.5 Forecasting Using Machine Learning.

5. Statistical Framework

5.1 Statistical Hypotheses

The statistical hypotheses guiding this study were presented in Section 2.4 in relation to the four research questions. The hypotheses provide a formal framework for examining the associations between predictor variables and one-hour-ahead PM2.5 concentration, the predictive performance of the machine learning models, differences among the evaluated models, and the contribution of predictor variables to the selected model.

The hypotheses were specified using a conventional 5% significance framework ($\alpha = 0.05$). Because the primary purpose of the study was predictive modelling, empirical

evaluation was based primarily on MAE, RMSE, R^2 , chronological train-test evaluation, and expanding-window time-series cross-validation.

5.2 Sample Size Calculation

Although the Beijing Multi-Site Air Quality Dataset contains 420,768 hourly observations, a minimum sample size calculation was performed to provide a conventional minimum-sample benchmark for the statistical framework of the study. Following the recommendation of Cochran (1977), the required sample size for a large population was estimated using:

$$n = \frac{Z^2 \times p(1 - p)}{e^2}$$

$$n = \frac{(1.96)^2 \times 0.50(1 - 0.50)}{(0.05)^2}$$

$$n = 384.16 \approx 385 \text{ observations}$$

where n_0 represents the minimum required sample size, Z is the Z-value corresponding to the desired confidence level (1.96 for a 95% confidence level), p is the estimated population proportion (0.50), and e is the margin of error (0.05).

5.3 Sample Size Justification

Following data preprocessing and construction of the one-hour-ahead forecasting target, the final analytical dataset contained 409,158 observations. This substantially exceeded the minimum sample size of 385 observations calculated using the Cochran formula. The available dataset therefore provided a substantially larger observational base than the minimum sample size used for the statistical justification.

The large dataset also provided extensive coverage of temporal and spatial variation in PM_{2.5} concentrations across the 12 monitoring stations. The number of available observations supported the implementation of the chronological train-test evaluation and five-fold expanding-window time-series cross-validation used in the study.

5.4 Significance Level

A significance level of $\alpha = 0.05$ was adopted for the statistical hypotheses, corresponding to a 95% confidence level. This significance level was specified as the decision criterion for interpreting the hypotheses presented in Section 2.4.

Because the primary objective of this study was predictive forecasting, model selection was based primarily on MAE, RMSE, and R^2 , together with chronological testing and expanding-window time-series cross-validation. The significance level therefore provides the statistical framework for the study's hypotheses but does not replace the predictive evaluation criteria used for model comparison.

5.5 Chapter Summary

This chapter established the hypotheses, minimum sample-size calculation, significance level, and statistical evaluation framework. The available dataset substantially exceeded the calculated minimum sample size of 385 observations, while MAE, RMSE, R^2 , chronological testing, and expanding-window validation provided the primary predictive evaluation framework.

6. Model Development and Evaluation

6.1 Introduction

This chapter describes the development and evaluation of the supervised machine learning regression models used to forecast one-hour-ahead PM_{2.5} concentration. The modelling process included preparation of the predictor variables, chronological train-test splitting, implementation of eight regression algorithms, comparative evaluation using MAE, RMSE, and R^2 , and temporal validation of the selected model.

The selected model was subsequently evaluated using expanding-window time-series cross-validation to assess temporal robustness and interpreted using gain-based feature importance and SHAP to identify influential predictors.

6.2 Predictor Variables

The predictor variables used in this study represented information available at the current forecasting hour. They included current-hour air-pollutant measurements (PM_{2.5}, PM₁₀, SO₂, NO₂, CO, and O₃), meteorological variables (TEMP, PRES, DEWP, RAIN, WSPM, and wind direction), temporal variables (year, month, day, and hour), and monitoring-station information.

The engineered target variable, PM_{2.5}_next_hour, represented the PM_{2.5} concentration one hour ahead within the same monitoring station. Therefore, current-hour PM_{2.5} was retained as a predictor rather than being used as the target.

After categorical encoding and predictor preparation, the modelling dataset contained 41 predictor variables. The record identifier (No) was excluded because it did not provide meaningful predictive information.

6.3 Chronological Train-Test Split

Because the dataset represents a time-dependent forecasting problem, a chronological 80:20 train-test split was used rather than random sampling. The earlier observations were used for model training, while the subsequent observations were reserved as an unseen chronological testing dataset.

Preserving temporal ordering reduced the risk of information leakage from future observations into the training data and provided a more realistic assessment of how the models would perform when forecasting future PM_{2.5} concentrations.

The same training and testing framework was applied consistently across all eight regression models to enable a fair comparison of their predictive performance.

6.4 Machine Learning Models

This section briefly introduces each model.

Model	Purpose
Linear Regression	Baseline linear model used to establish a reference level of predictive performance.
Decision Tree	Tree-based model used to capture nonlinear relationships and interactions among predictors.
Random Forest	Ensemble bagging model that combines multiple decision trees to improve predictive stability and reduce overfitting.
Gradient Boosting	Sequential boosting model that builds trees iteratively to reduce prediction errors.
AdaBoost	Adaptive boosting model that sequentially emphasizes observations that are difficult to predict.
XGBoost	Regularized gradient-boosting model designed to provide accurate and efficient predictions.
LightGBM	Histogram-based gradient-boosting model designed for efficient learning from large structured datasets.
CatBoost	Gradient-boosting model designed to handle categorical variables efficiently while capturing nonlinear relationships.

The models were selected to provide a diverse comparison of linear, tree-based, bagging, and boosting approaches. Applying the same predictor set and evaluation framework to all eight models enabled the study to identify the model providing the strongest predictive performance for one-hour-ahead PM_{2.5} forecasting.

6.5 Model Evaluation Metrics

Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

- **Mean Absolute Error (MAE)**

Mean Absolute Error measures the average absolute difference between the observed and predicted PM2.5 concentrations. Lower MAE values indicate better predictive accuracy because they represent smaller average prediction errors. In the context of air quality forecasting, a lower MAE indicates more accurate short-term PM2.5 predictions, enabling more reliable air quality advisories and timely public health interventions.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- **Root Mean Squared Error (RMSE)**

Root Mean Squared Error measures the square root of the average squared prediction errors. Because larger errors receive greater weight, RMSE is particularly useful for assessing the impact of substantial forecasting errors. In environmental forecasting, a lower RMSE reduces the likelihood of large prediction errors that could affect air quality warnings and environmental decision-making.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- **Coefficient of Determination (R^2)**

The coefficient of determination (R^2) measures the proportion of variance in the observed PM2.5 concentrations explained by the regression model. Higher R^2 values indicate that the model explains a greater proportion of the variation in the observed target values, with a value of 1 representing perfect prediction. Therefore, higher R^2 values generally indicate better predictive performance.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

where: y_i = observed PM2.5 concentration; $\hat{y}_i$ = predicted PM2.5 concentration; $\bar{y}$ = mean observed PM2.5 concentration; n = number of observations

6.6 Expanding-Window Time-Series Cross-Validation

To evaluate the temporal robustness of the selected forecasting model, five-fold expanding-window time-series cross-validation was performed. Unlike random cross-validation, the expanding-window approach preserves the chronological ordering of observations by progressively increasing the size of the training set while evaluating performance on subsequent unseen observations. This evaluation strategy provides a more realistic assessment of forecasting performance for time-dependent data (Bergmeir et al., 2018).

6.7 Model Interpretability

Model interpretability was assessed using gain-based feature importance and SHAP (SHapley Additive exPlanations) for the selected LightGBM model. Gain-based feature importance was used to examine the relative contribution of predictor variables to the model's tree-based learning process. SHAP analysis was used to quantify the contribution of individual predictor variables to the model's predictions and to examine the direction and magnitude of their effects (Lundberg & Lee, 2017).

The combination of gain-based feature importance and SHAP provided complementary information regarding the predictors contributing to the one-hour-ahead PM2.5 forecasts. This interpretability analysis supported the identification of the most influential predictive variables while emphasizing that feature contributions represent model associations rather than causal effects.

6.8 Chapter Summary

This chapter described predictor preparation, chronological train-test splitting, implementation of eight regression algorithms, evaluation using MAE, RMSE, and R^2 , and five-fold expanding-window validation. LightGBM was selected as the leading model and subsequently interpreted using gain-based feature importance and SHAP.

The following chapter presents the experimental results and findings, including comparative model performance, cross-validation results, feature-importance findings, SHAP analysis, and the results associated with the study's research questions.

7. Results and Discussion

7.1 Introduction

This chapter presents comparative model performance, expanding-window validation, feature-importance and SHAP findings, and their interpretation in relation to the four research questions and practical implications.

7.2 Comparative Model Performance

The eight supervised machine learning regression models were evaluated on the training and chronological testing datasets using MAE, RMSE, and R^2 . The comparative results are presented in Table 7.2.

Table 7.2

Comparative Performance of Supervised Machine Learning Regression Models in the Chronological Train-Testing Dataset

Model	Train MAE	Train RMSE	Train R^2	Test MAE	Test RMSE	Test R^2
LightGBM	9.77	17.33	0.9531	9.67	18.26	0.9522
XGBoost	9.70	17.11	0.9543	9.77	18.57	0.9506

Model	Train MAE	Train RMSE	Train R ²	Test MAE	Test RMSE	Test R ²
Gradient Boosting	10.25	18.66	0.9457	9.78	18.64	0.9502
Random Forest	3.69	6.83	0.9927	10.16	19.02	0.9482
Linear Regression	10.76	19.63	0.9399	10.17	19.08	0.9478
CatBoost	10.36	18.87	0.9444	10.07	19.41	0.9460
Decision Tree	0.00	0.00	1.0000	16.13	29.14	0.8783
AdaBoost	21.43	29.01	0.8687	21.47	29.45	0.8757

Table 7.2 shows clear differences in predictive performance across the eight models. Decision Tree exhibited pronounced overfitting, with training $R^2 = 1.0000$ compared with testing $R^2 = 0.8783$, while Random Forest showed a smaller training-testing gap (0.9927 vs. 0.9482).

LightGBM, XGBoost, and Gradient Boosting demonstrated the strongest testing performance. LightGBM achieved the lowest testing MAE ($9.67 \mu\text{g}/\text{m}^3$), lowest RMSE ($18.26 \mu\text{g}/\text{m}^3$), and highest R^2 (0.9522), followed by XGBoost ($R^2 = 0.9506$) and Gradient Boosting ($R^2 = 0.9502$).

LightGBM showed limited evidence of overfitting, with training $R^2 = 0.9531$ and testing $R^2 = 0.9522$. The corresponding RMSE values were 17.33 and $18.26 \mu\text{g}/\text{m}^3$, indicating that performance remained stable on unseen chronological observations.

Linear Regression and CatBoost also performed strongly, whereas AdaBoost and Decision Tree produced substantially higher testing errors and lower R^2 values.

Overall, LightGBM provided the strongest predictive performance across all three-testing metrics and was therefore selected as the final forecasting model for expanding-window time-series cross-validation and subsequent interpretability analysis using gain-based feature importance and SHAP.

7.3 Selection of the Best-Performing Model

LightGBM was selected because it achieved the best testing MAE, RMSE, and R^2 , with closely aligned training and testing R^2 values, indicating stable performance on unseen chronological observations.

7.4 Expanding-Window Time-Series Cross-Validation

Five-fold expanding-window time-series cross-validation was performed using the selected LightGBM model to assess temporal robustness. The method progressively increased the training dataset while evaluating performance on subsequent unseen observations, preserving chronological order and reducing the risk of temporal leakage (Bergmeir et al., 2018).

Table 7.4

Five-Fold Expanding-Window Time-Series Cross-Validation Results for the Selected LightGBM Model

Fold	Training Observations	Validation Observations	MAE ($\mu\text{g}/\text{m}^3$)	RMSE ($\mu\text{g}/\text{m}^3$)	R^2
1	54,722	55,272	20.32	38.56	0.8258
2	109,994	54,259	9.37	15.99	0.9235
3	164,253	54,294	10.92	19.87	0.9498
4	218,547	54,199	7.83	13.40	0.9418
5	272,746	54,579	11.49	24.33	0.9422
Average	—	—	11.98	22.43	0.9166

Table 7.4 shows that LightGBM maintained strong predictive performance across the five validation periods. The mean MAE was $11.98 \mu\text{g}/\text{m}^3$, mean RMSE was $22.43 \mu\text{g}/\text{m}^3$, and mean R^2 was 0.9166, indicating substantial predictive capability across different chronological periods.

Performance varied across folds. Fold 1 produced the weakest results (MAE = $20.32 \mu\text{g}/\text{m}^3$; RMSE = $38.56 \mu\text{g}/\text{m}^3$; $R^2 = 0.8258$), whereas Fold 4 achieved the lowest MAE ($7.83 \mu\text{g}/\text{m}^3$) and RMSE ($13.40 \mu\text{g}/\text{m}^3$), and Fold 3 achieved the highest R^2 (0.9498). The variation indicates that forecasting difficulty differed across historical periods.

The standard deviations were 4.8721 for MAE, 9.9146 for RMSE, and 0.0517 for R^2 . Despite the variation, R^2 exceeded 0.92 in four of the five folds, supporting the temporal robustness of LightGBM for one-hour-ahead PM2.5 forecasting.

7.5 Model Interpretability

Interpretability analysis of the selected LightGBM model used gain-based feature importance and SHAP (SHapley Additive exPlanations) to identify the most influential predictors and their contributions to one-hour-ahead PM2.5 forecasts.

7.5.1 Gain-Based Feature Importance

Gain-based feature importance identified current-hour PM2.5 as the dominant predictor, accounting for 98.17% of total gain. PM10 (0.43%), wind speed (WSPM; 0.36%), hour of the day (0.19%), dew point temperature (DEWP; 0.16%), and temperature (TEMP; 0.13%) followed, while all remaining predictors contributed less than 0.10% each. Figure 7.5.1 presents the top 15 predictors based on gain-based feature importance.

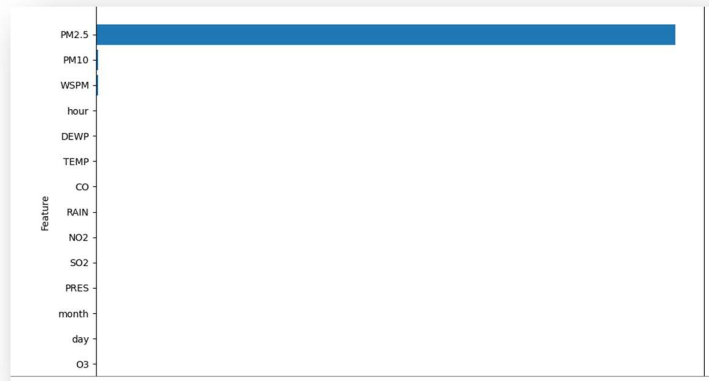


Figure 7.5.1

Top 15 Feature Importances for the Selected LightGBM Model

Table 7.5.1

Top 15 Predictor Variables Based on Gain-Based Feature Importance

Rank	Predictor	Gain	Importance (%)
1	PM2.5	1.03E+10	98.17
2	PM10	4.55E+07	0.43
3	WSPM	3.77E+07	0.36
4	Hour	2.05E+07	0.19
5	DEWP	1.63E+07	0.16
6	TEMP	1.32E+07	0.13
7	CO	1.09E+07	0.1
8	RAIN	9.25E+06	0.09
9	NO ₂	7.03E+06	0.07
10	SO ₂	6.69E+06	0.06
11	PRES	4.76E+06	0.05
12	Month	4.68E+06	0.04

Rank	Predictor	Gain	Importance (%)
13	Day	4.46E+06	0.04
14	O ₃	2.65E+06	0.03
15	Wind Direction (NW)	1.17E+06	0.01

Note. Gain represents the contribution of each predictor to the LightGBM model based on the improvement in the model's objective function resulting from splits involving the predictor. Importance (%) represents the relative contribution based on gain.

7.5.2 SHAP (SHapley Additive exPlanations) Analysis

SHAP (SHapley Additive exPlanations) analysis was performed to examine how individual predictor variables contributed to the one-hour-ahead PM_{2.5} predictions generated by the selected LightGBM model. The SHAP summary plot provides both the relative importance of predictors and the direction of their contribution to model predictions.

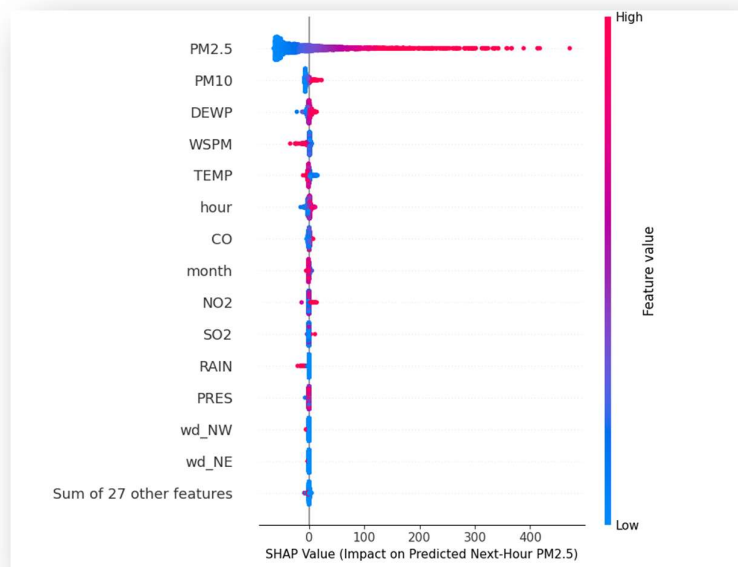


Figure 7.5.2

SHAP Summary Plot for One-Hour-Ahead PM_{2.5} Forecasting

As shown in Figure 7.5.2, PM2.5 was by far the most influential predictor. Higher current-hour PM2.5 values generally contributed positively to the following-hour prediction, while lower values contributed negatively. PM10, DEWP, WSPM, and TEMP followed in importance, although their contributions were substantially smaller. The remaining predictors showed comparatively limited SHAP contributions.

The SHAP results were broadly consistent with the gain-based feature-importance results, with current-hour PM2.5 clearly dominating both analyses. The results also indicate that meteorological and temporal variables provided additional predictive information and that their contributions varied across observations, reflecting nonlinear relationships captured by LightGBM.

7.6 Research Question-wise Findings

The findings were examined in relation to the four research questions established in Chapter 2. The results from model evaluation, expanding-window cross-validation, gain-based feature importance, and SHAP analysis provide evidence for addressing each research question.

Research Question 1 (RQ1):

Which current-hour pollutant and meteorological variables are most strongly associated with PM2.5 concentration one hour ahead?

The feature-importance and SHAP analyses identified current-hour PM2.5 as the dominant predictor of one-hour-ahead PM2.5 concentration. Other important predictors included PM10, wind speed (WSPM), dew point temperature (DEWP), temperature (TEMP), and hour of the day. These findings indicate that recent pollution levels provide the strongest predictive information, while meteorological and temporal variables provide additional information for short-term forecasting.

Research Question 2 (RQ2):

How accurately can machine learning regression models forecast PM2.5 concentration one hour ahead using current-hour pollutant, meteorological, temporal, wind-direction, and monitoring-station information?

The eight evaluated regression models demonstrated strong predictive capability on the chronological testing dataset. LightGBM achieved the lowest test MAE (9.67 $\mu\text{g}/\text{m}^3$), lowest test RMSE (18.26 $\mu\text{g}/\text{m}^3$), and highest test R^2 (0.9522) among the evaluated models. These results indicate that the models were able to explain a substantial proportion of the variation in one-hour-ahead PM2.5 concentration

Research Question 3 (RQ3):

Which machine learning regression model provides the best predictive performance for one-hour-ahead PM2.5 forecasting?

LightGBM was the best-performing model, achieving the lowest MAE and RMSE and highest R^2 among the eight evaluated models. Its temporal robustness was further supported by five-fold expanding-window cross-validation, with mean MAE = 11.98 $\mu\text{g}/\text{m}^3$, mean RMSE = 22.43 $\mu\text{g}/\text{m}^3$, and mean $R^2 = 0.9166$.

Research Question 4 (RQ4):

Which predictor variables contribute most strongly to the forecasts of the selected machine learning model?

Both gain-based feature importance and SHAP analysis identified current-hour PM2.5 as the most influential predictor. The gain-based analysis attributed 98.17% of total feature importance to current-hour PM2.5, followed by PM10, WSPM, hour, DEWP, and TEMP. SHAP analysis similarly showed substantially greater influence from PM2.5 than from the remaining predictors, while meteorological and temporal variables provided additional predictive information.

7.7 Practical Implications

The findings demonstrate the practical potential of the proposed LightGBM framework for short-term PM_{2.5} forecasting using routinely collected pollutant, meteorological, temporal, and monitoring-station information. The framework could support environmental monitoring agencies and air quality management authorities by generating one-hour-ahead PM_{2.5} estimates to identify potentially deteriorating air quality and support timely monitoring or advisory preparation.

The forecasts could also provide an additional information source for public health communication, particularly during periods of elevated pollution. However, the model is intended as a decision-support tool rather than a replacement for official air quality monitoring or health guidance. The use of LightGBM together with SHAP provides both predictive capability and information about the main variables contributing to individual forecasts.

Figure 7.7 illustrates a proposed operational framework in which current-hour observations pass through the established preprocessing and feature-engineering pipeline before being provided to the trained LightGBM model. The resulting forecast could be displayed through a monitoring dashboard or alerting system. For example, observations received from a station at 2:00 PM could generate a 3:00 PM PM_{2.5} forecast, allowing monitoring personnel to identify potentially elevated conditions and prioritize further monitoring or advisory preparation. This represents a proposed implementation rather than a production deployment and would require live-data integration, system testing, monitoring, and model maintenance.

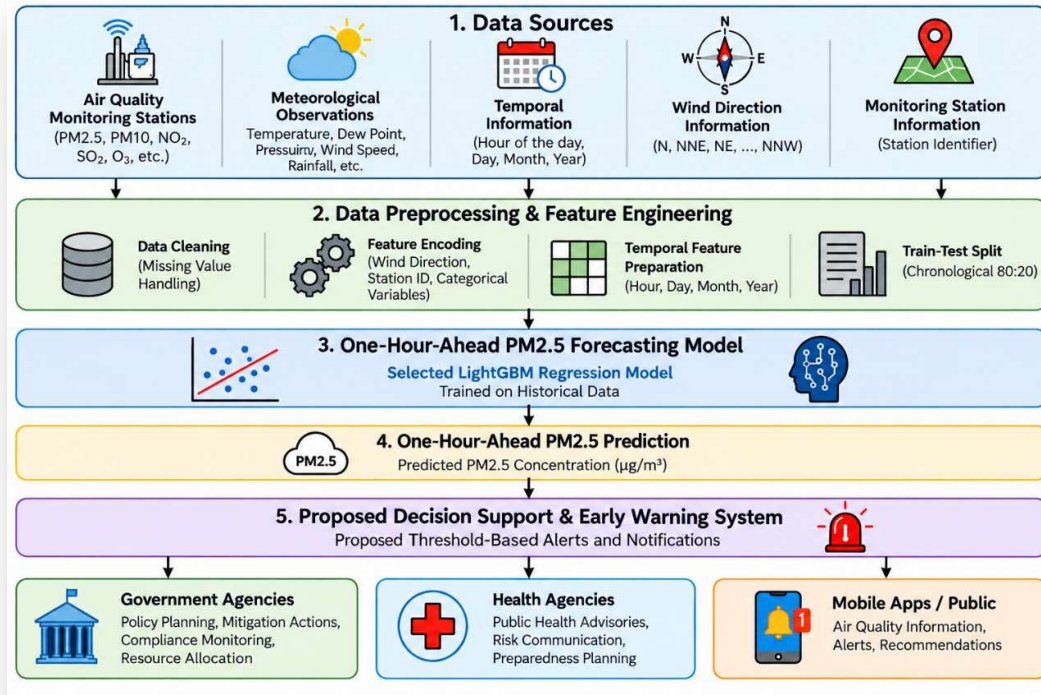


Figure 7.7

Proposed Deployment Framework for One-Hour-Ahead PM2.5 Forecasting Using the Selected LightGBM Model

7.8 Chapter Summary

LightGBM demonstrated the strongest chronological testing performance and substantial predictive capability across five-fold expanding-window validation. Gain-based feature importance and SHAP identified current-hour PM2.5 as the dominant predictor, supporting the potential use of the framework for short-term air-quality monitoring and decision support.

8. Conclusion and Future Work

8.1 Summary of the Study

This study developed and evaluated a supervised machine learning framework for one-hour-ahead PM2.5 forecasting using the Beijing Multi-Site Air Quality Dataset. Current-

hour pollutant concentrations, meteorological conditions, temporal variables, wind-direction information, and monitoring-station characteristics were used as predictors. Eight supervised regression models were evaluated using a chronological train-test framework, with performance assessed using MAE, RMSE, and R^2 .

LightGBM demonstrated the strongest predictive performance, achieving a test MAE of $9.67 \mu\text{g}/\text{m}^3$, RMSE of $18.26 \mu\text{g}/\text{m}^3$, and R^2 of 0.9522. Five-fold expanding-window validation produced mean MAE of $11.98 \mu\text{g}/\text{m}^3$, RMSE of $22.43 \mu\text{g}/\text{m}^3$, and R^2 of 0.9166. Gain-based feature importance and SHAP analysis identified current-hour PM_{2.5} as the dominant predictor, with additional information provided by pollutant, meteorological, and temporal variables.

8.2 Key Findings

The principal findings of the study are:

- **LightGBM achieved the strongest predictive performance** among the eight evaluated models.
- **Five-fold expanding-window validation supported temporal robustness**, with substantial predictive capability across different chronological periods.
- **Current-hour PM_{2.5} was the dominant predictor**, accounting for 98.17% of gain-based feature importance.
- **PM₁₀, WSPM, hour, DEWP, and TEMP** provided additional predictive information.
- **SHAP analysis supported the gain-based findings** and provided information about the direction and magnitude of predictor contributions.
- The findings collectively address the four research questions and support the potential application of the proposed framework for short-term PM_{2.5} forecasting

8.3 Contributions of the Study

The study contributes a systematic comparison of eight supervised regression models, combines chronological testing with expanding-window validation, and integrates gain-based feature importance with SHAP for model interpretability.

8.4 Limitations

The findings of this study should be interpreted in consideration of several limitations.

First, the analysis was conducted using the Beijing Multi-Site Air Quality Dataset, which contains observations from monitoring stations located in Beijing, China. Consequently, the findings may not generalize directly to other geographical regions with different climatic conditions, emission sources, and pollution characteristics.

Second, the forecasting framework was limited to one-hour-ahead PM_{2.5} prediction. Although this forecasting horizon is relevant for short-term air quality monitoring, longer forecasting horizons may involve additional temporal dependencies and forecasting challenges that were beyond the scope of this study.

Third, the analysis was restricted to the pollutant, meteorological, temporal, wind-direction, and monitoring-station variables available in the dataset. Potentially relevant external factors, such as traffic intensity, industrial emissions, land-use characteristics, satellite observations, and other emission-related information, were not included.

Finally, the study evaluated eight conventional supervised machine learning regression algorithms and did not compare them with deep learning or other advanced time-series architectures. Models such as LSTM, Temporal Convolutional Networks, and Transformer-based approaches were therefore outside the scope of the present investigation.

8.5 Future Research

Future research could evaluate the framework across multiple cities and geographical regions and examine longer forecasting horizons such as 6, 12, and 24 hours to assess generalizability and temporal performance.

Additional contextual data, including satellite observations, traffic, industrial emissions, land-use characteristics, and weather forecasts, could be incorporated to improve prediction.

Future studies could also compare LightGBM with LSTM, TCN, and Transformer-based models, considering predictive accuracy, computational efficiency, temporal robustness, and interpretability.

8.6 Final Conclusion

This study demonstrated the potential of supervised machine learning, particularly LightGBM, for one-hour-ahead PM_{2.5} forecasting using routinely collected pollutant, meteorological, temporal, wind-direction, and monitoring-station information. LightGBM achieved the strongest predictive performance among the evaluated models, while expanding-window validation supported its predictive capability across different chronological periods.

The interpretability analysis identified current-hour PM_{2.5} as the dominant predictor, with SHAP analysis providing complementary evidence regarding predictor contributions. Overall, the proposed framework combines predictive performance, temporal validation, and interpretability and has potential to support urban air quality monitoring and decision-support applications, subject to further validation across locations and operational settings.

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